

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing** *Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed and implemented a scalable AI chatbot prototype leveraging large language models to transform customer service interactions. Utilized retrieval-augmented generation (RAG) architecture combined with vector databases to embed and query company knowledge bases. Experimented with embedding techniques to optimize retrieval performance and integrated OpenAI API for real-time response generation, significantly reducing human agent workload by 35% and achieving instant query responses. Collaborated closely with product teams to deploy the solution as a production-ready AI system, bridging research innovations with practical application.

Tech: Python, LangChain, OpenAI API, ChromaDB, RAG Architecture, LLM Integration

Predictive Maintenance System for Industrial Machinery (2024)

Conducted applied research on sensor data to develop a machine learning-based predictive maintenance model for manufacturing equipment. Extracted features from large-scale industrial IoT time-series sensor data and implemented Random Forest classifiers to forecast machine failures. Evaluated model performance rigorously using F1-score and precision metrics. Translated research findings into a deployable pipeline that reduced unexpected downtime by 28%, demonstrating strong capability in transforming research concepts into production-ready solutions that optimize operational efficiency.

Tech: Python, Random Forest, Time-Series Analysis, Sensor Data Processing

Customer Sentiment Classification using NLP (2024)

Researched and built a natural language processing pipeline that analyzed customer reviews to classify sentiment and extract actionable insights. Employed advanced preprocessing techniques including tokenization, stopword removal, and TF-IDF vectorization. Designed and implemented logistic regression models to evaluate new approaches for sentiment classification achieving 89% accuracy. Applied research insights to propose enhancements in product quality, effectively bridging theoretical NLP advances with business impact.

Tech: Python, NLTK, TF-IDF, Logistic Regression

Sales Forecasting Model for Retail Inventory Optimization (2024)

Developed and prototyped time series forecasting models to predict monthly retail sales, facilitating strategic inventory planning and revenue optimization. Conducted in-depth time-series decomposition and seasonality analysis, implementing and comparing ARIMA and Facebook Prophet models. Validated model performance with MAE and RMSE metrics, achieving a 25% reduction in forecasting error. Integrated research methodologies into deployable forecasting pipelines, thereby supporting data-driven decision-making and reducing stockouts.

Tech: Python, Prophet, ARIMA, Pandas, Time Series Analysis

About :

I specialize in bridging cutting-edge research and real-world applications by designing, training, and deploying scalable, production-ready AI and ML models. Proficient in Python and ML frameworks such as TensorFlow and PyTorch, I excel at implementing and evaluating novel machine learning and deep learning algorithms. Experienced in developing end-to-end ML pipelines including data preprocessing, feature engineering, model optimization, and large dataset experimentation. Skilled at translating research ideas and academic papers into deployable engineering solutions, collaborating effectively with AI engineers, data scientists, and product teams. Committed to staying up to date with the latest research and industry advancements to build efficient, reliable, and impactful AI systems.

